PRADYUMNA PRADHAN

Contact Information

SAMPL Lab, Kimmel Building, +91-9692581139
Department of Computer Science and Applied Mathematics, pradpradyumna@gmail.com
Weizmann Institute of Science, Rehovot, 7610001, Israel. pradyumna.pradhan@weizmann.ac.il

Research Interests

Model-based Ultrasound Imaging, Compressive Sensing, Large Dataset Compression and reconstruction, Coreset Generation, Neural Network Compression, Deep Unfolded Networks.

Education

Weizmann Institute of Science, Rehovot, Israel
Postdoctoral fellow in the Department of Computer Science and Applied Mathematics *Nov 2026 - Present*
Advisor: Prof. Yonina Eldar

Indian Institute of Petroleum and Energy, Visakhapatnam, India
PhD in Mathematics *July 2021 - October 2026*
Advisor: Dr. Ramunaidu Randhi

National Institute of Technology, Rourkela, India
Integrated M.Sc. (B.Sc. + M.Sc.) in Mathematics *July 2015-May 2020*
CGPA: **9.18/10**; Advisor: Dr. Divya Singh

Gayatri Junior College, Berhampur, India
Higher Secondary Education (Grade 12) *July 2013-April 2015*
Percentage: **82.67%**

S. D. B. High School, Barida(A), India
Board of Secondary Education (Grade 10) *June 2013*
Percentage: **79.83%**

Journal Publication

S. M. A. Anwar, **P. Pradhan**, P. Sasmal, and R. Randhi, "Bound on RIP of order s+1 for HTP Algorithm," Accepted in *IEEE Signal Processing Letters*, 2026.

P. Pradhan, P. Sasmal, and R. Randhi, "A Frame-Theoretic Approach to Robust Filter Pruning in Convolutional Neural Networks," Accepted in *IEEE Signal Processing Letters*, 2026.

Y. Kvich, P. Reshma, **P. Pradhan**, R. Randhi, and Y. C. Eldar, "Deep Unfolding of Tail-Based Methods for Robust Sparse Recovery," *IEEE Transactions on Neural Networks and Learning Systems*, 2025.

P. Pradhan, S. K. Jain, P. Sasmal, and R. Randhi, "Non-iterative Sparse Signal Recovery Algorithms for Sparse Matrices," *Journal of the Franklin Institute*, 2025.

P. Pradhan, S. M. A. Anwar, R. Randhi, and P. Sasmal, "Non-negative Sparse Signal Recovery using ReLU and Hard Thresholding Pursuit," *Signal Processing*, 2025.

P. Pradhan, S. K. Jain, P. Sasmal, and R. Randhi, "Deterministic Construction of Unimodular Tight Frames Consisting Orthogonal Blocks," *Signal Processing*, 2025.

S. M. A. Anwar, **P. Pradhan**, P. Sasmal, and R. Randhi, "Sufficient Conditions for Convergence of RHT and RHTP Algorithms," *IEEE Signal Processing Letters*, 2025.

B. S. Shaik, **P. Pradhan**, W. Pu, R. Randhi, M. R. D. Rodrigues, and Y. C. Eldar, "Optimization Guarantees of Unfolded ISTA and ADMM Networks with Smooth Soft Thresholding," *IEEE Transactions on Signal Processing*, 2024.

Submitted/ Under Preparation

P. Pradhan, P. Sasmal, R. Randhi, and Y. C. Eldar, "Filter Pruning for Convolutional Neural Networks via Non-negative Sparse Recovery," 2026.

S. M. A. Anwar, **P. Pradhan**, P. Sasmal, and R. Randhi,"Improved $s+1$ order RIC Bound for the IHT Algorithm," 2026.

S. K. Jain, **P. Pradhan**, R. Randhi, and P. Sasmal, "Block-Disjunct Algorithm: A Fast, Non-iterative Framework for Block-Sparse Recovery with Sparse Matrices," 2026.

S. K. Jain, **P. Pradhan**, P. Sasmal, and R. Randhi, "Block-Sparse Recovery via Disjunct Property under High Block Coherence," 2026.

S. K. Jain, **P. Pradhan**, R. Randhi, and P. Sasmal, "Disjunct-Block-OMP: A Fast Block Sparse Signal Recovery Algorithm for Embedded Matrices," 2025.

P. Pradhan, S. Mukherjee, R. Randhi, and P. Sasmal, "Compressed Sensing-Based Model and Data Compression for Efficient Training and Inference in Chest X-Ray Imaging," (Under Preparation).

Conference Publication

P. Pradhan, S. Mukherjee, R. Randhi, and P. Sasmal, "Efficient Coreset Generation for Chest X-ray Imaging Using Compressed Sensing," Accepted in *IEEE ICIP*, 2026.

P. Reshma, **P. Pradhan**, H. Singh, S. Tenneti, P. Sasmal, and R. Randhi, "Deep Learned-CoSaMP Network for Signal Denoising," *IEEE SampTA*, 2025.

A. Sen, **P. Pradhan**, R. Randhi, and C. S. Sastry, "Learned ReLU-based Soft Thresholding: A data-driven method for non-negative sparse signal recovery," *IEEE ICASSP*, 2025.

A. Sen, **P. Pradhan**, R. Randhi, and C. S. Sastry, "Unrolled Proximal Gradient Descent Method for Non-Negative Least Squares Problem," *IEEE ICASSP*, 2024.

P. Pradhan, S. B. Shah, R. Randhi and Y. C. Eldar, "Recursive-Tail-Fista for Sparse Signal Recovery," *IEEE ICASSP*, 2024.

Technical Skills

Programming: Python, MATLAB, C/C++
Software & Tools: PyTorch, CUDA, Jupyter Notebook, Google Colab, LaTeX

Professional Activities

Presented research at reputed conferences, including RAAM, ICASSP, and SampTA.

Communicated technical research to interdisciplinary academic audiences.

Served as a reviewer for peer-reviewed journals, including IEEE TSP, Signal Processing, and Signal, Image and Video Processing.

Achievements

Awarded the International Travel Scheme (ITS) Grant by the Anusandhan National Research Foundation (ANRF), Government of India, for participation in SampTA 2025.

Recipient of the IEEE Signal Processing Society Travel Grant for participation in ICASSP 2024.

Qualified the Graduate Aptitude Test in Engineering (GATE) 2021 in Mathematics discipline.

Recipient of the Academic Excellence Award for securing First Rank in the Department of Mathematics for five consecutive years (2015–2020) at NIT Rourkela.

References

Dr. Ramunaidu Randhi
Associate Professor, Department of Humanities and Sciences (Mathematics)
Indian Institute of Petroleum and Energy, Visakhapatnam, India
Email: ramu.math@iipe.ac.in

Prof. C S Sastry
Professor, Department of Mathematics
Indian Institute of Technology Hyderabad, India
Email: csastry@iith.ac.in

Dr. Pradip Sasmal
Assistant Professor, Department of Mathematics
Indian Institute of Technology Jodhpur, India
Email: psasmal@iitj.ac.in